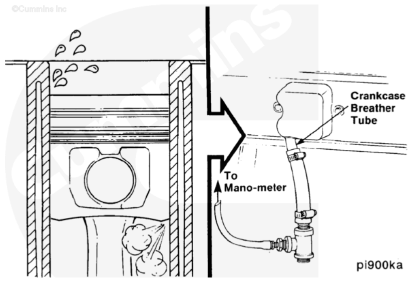
Trainings

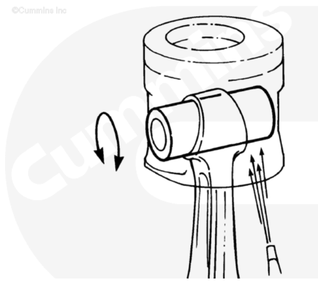
General Information

There are a number of power-related problems, including excessive lubricating oil consumption, smoke, blowby, and poor performance, that can be caused by inadequate sealing between the piston rings and the cylinder walls. A blowby measurement can help detect the problem. Refer to Procedure 014-010 in Section 14. (/qs3/pubsys2/xml/en/procedures/441/441-014-010.html)



pi900ka

A free-floating, hollow piston pin is used to attach the piston to the connecting rod. Lubricating the pin and journal is accomplished by residual spray from the piston cooling passage in the cylinder block.



pi9piga

Preparatory Steps

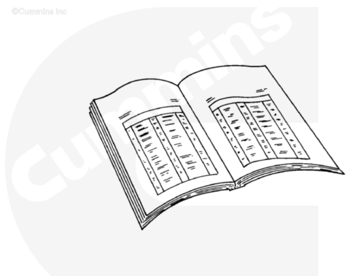
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠



ck800wa

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

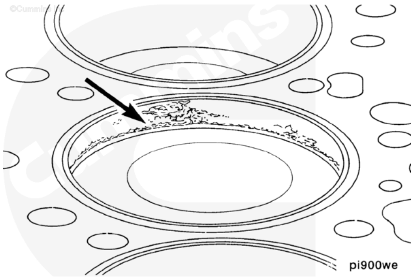
⚠ WARNING ⚠

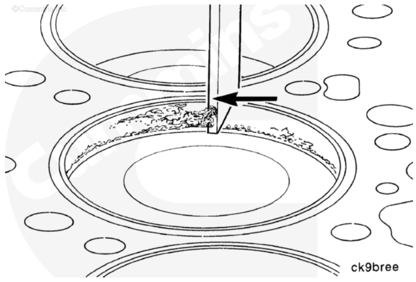
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/441/441-007-037.html)
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/441/441-008-018-tr.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/441/441-007-025.html)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/441/441-002-004.html)

Remove

Rotate the crankshaft with an engine barring tool until the pistons are below the carbon deposits, which are found above the ring travel area.

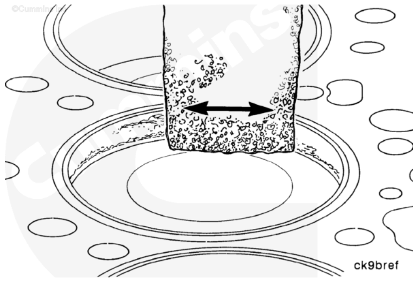




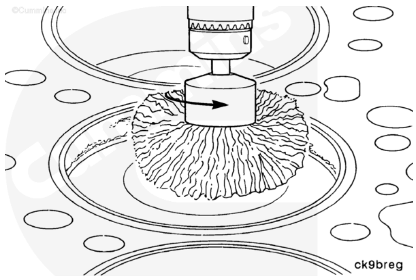
⚠ CAUTION ⚠

Do not use emery cloth or sandpaper to remove carbon from the cylinder bores. Aluminum oxide or silicon particles from these materials can cause serious engine damage.

Use a scraper or a blunt-edged instrument to loosen the carbon deposits. Do **not** damage the cylinder with the scraper.



Remove the remaining carbon deposits with an abrasive pad, Part Number 3823258, or equivalent.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

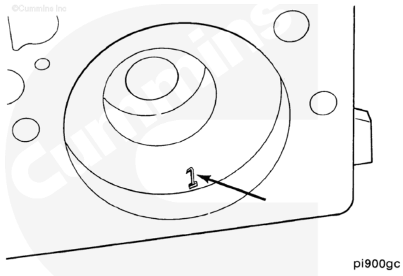
⚠ CAUTION ⚠

Do not use the steel wire wheel in the piston travel area. Operate the wheel in a circular motion to remove the deposits. An inferior quality wire wheel will lose steel bristles during operation, thus causing additional contamination.

An alternative method to remove the carbon ridge is to use a high-quality steel wire wheel installed in a drill or die grinder.

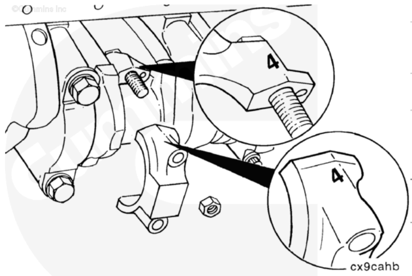
Mark each piston according to the cylinder location.

On pistons with anodized coatings, do **not** stamp the anodized coating on the outer rim.



Rotate the crankshaft to position the connecting rod caps at bottom dead center for removal.

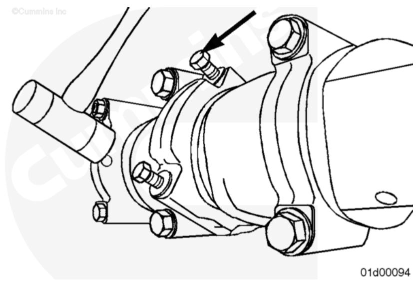
Mark each connecting rod and connecting rod cap according to the cylinder number location.



Note : Do not remove the capscrews from the connecting rods at this time.

Loosen the connecting rod capscrews.

Use a rubber hammer to hit the connecting rod capscrews to loosen the caps.

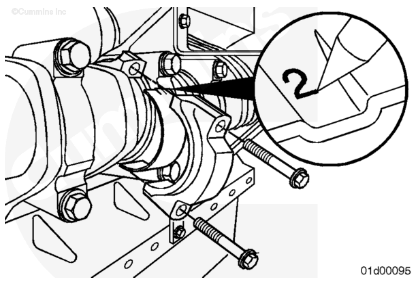


Remove the connecting rod capscrews.

Remove the connecting rod cap.

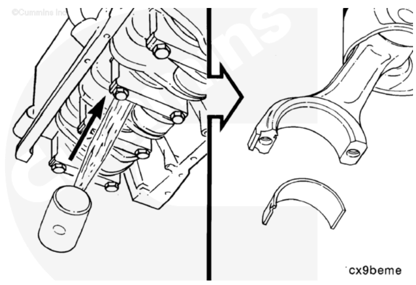
Remove the lower connecting rod bearing.

Mark the cylinder number and the letter “L” (lower) on the flat surface of the bearing tang.

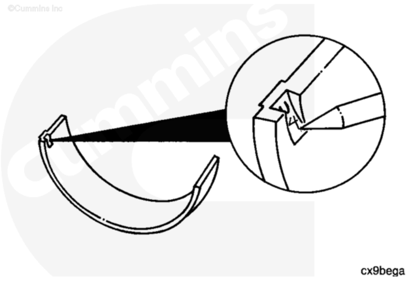


Push the connecting rod and piston assembly out of the cylinder bore. Care **must** be taken **not** to damage the connecting rod or bearing.

Remove the upper rod bearing.



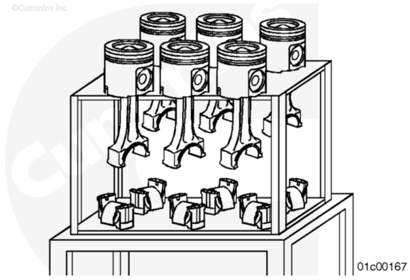
Mark the cylinder number and the letter “U” (upper) on the flat surface of the bearing tang.



The piston and connecting rod assemblies **must** be installed in the same cylinder number from which they were removed, to provide proper fit of worn mating surfaces, if parts are reused.

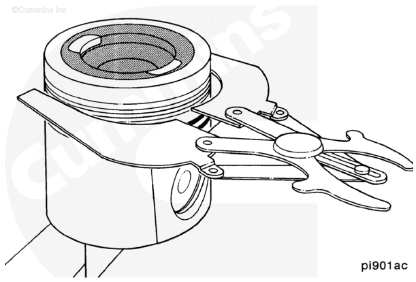
Use a tag to mark the cylinder number from which each piston and rod assembly were removed.

Place the rod and piston assemblies in a container designed to protect them from damage.

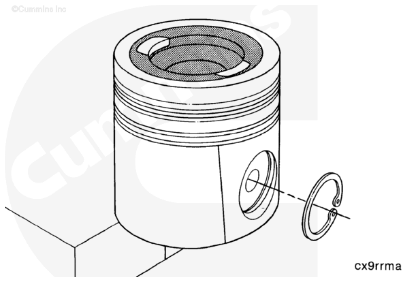


Disassemble

Use piston ring expander, Part Number 3823137, or equivalent, to remove the piston rings.



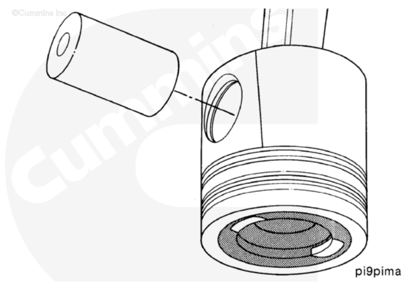
Remove the piston pin retaining rings.



Note : Heating the piston is not required.

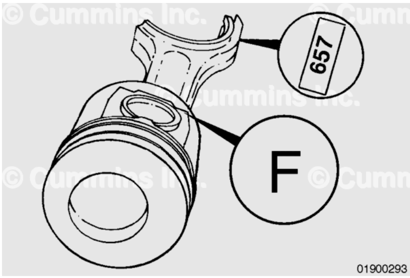
Remove the piston pin.

Remove the connecting rod from the piston.

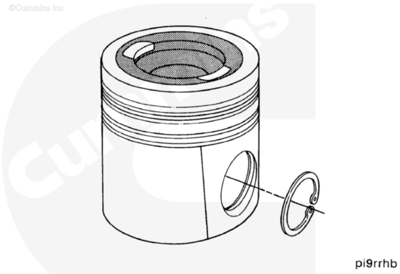


Assemble

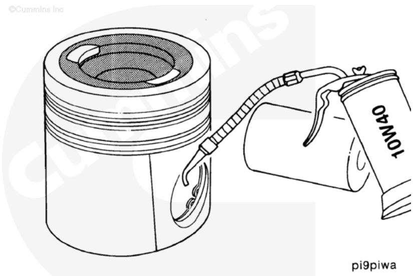
Make sure the “F” marking on the piston skirt and the numbers on the connecting rod and cap are oriented as illustrated. The piston bowl **must** be oriented toward the intake side of the engine.



Install the retaining ring in the pin groove on the front side of the piston.



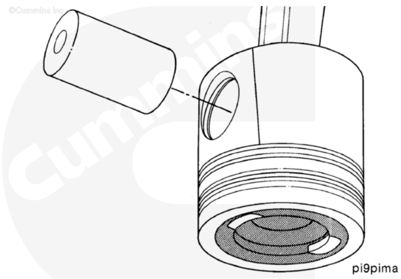
Lubricate the pin and pin bores with clean 15W-40 engine lubricating oil.



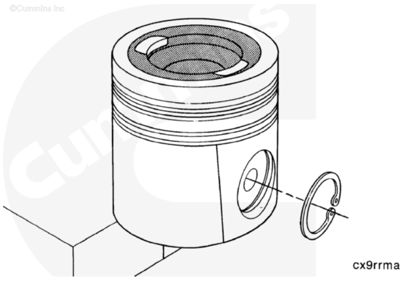
Note : Pistons do not require heating to install the pin; however, the pistons do need to be at room temperature or above.

Install the connecting rod.

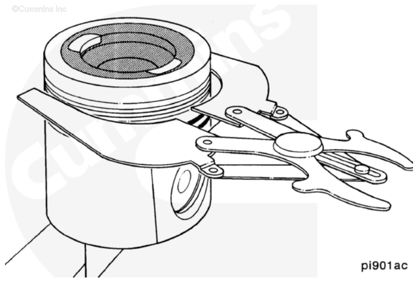
Install the piston pin.



Install the second retaining ring.



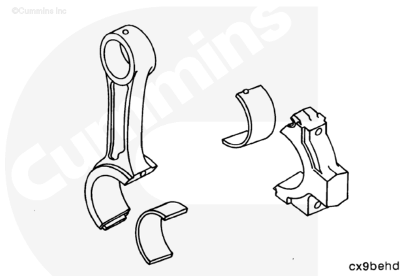
Use the piston ring expander, Part Number 3823137, or equivalent, to install the piston rings. Refer to Procedure 001-047 in Section 1. (/qs3/pubsys2/xml/en/procedures/441/441-001-047.html)



Install

⚠ CAUTION ⚠

The connecting rods and connecting rod caps are not interchangeable. The connecting rods and connecting rod caps are machined as an assembly. Failure will result if the connecting rods and caps are mixed.

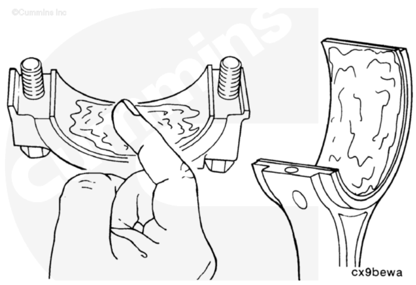


When rebuilding an engine with the original cylinder block, crankshaft, and pistons, make sure the pistons are installed in their original cylinders.

Install the bearing shells into both the connecting rod and connecting rod cap.

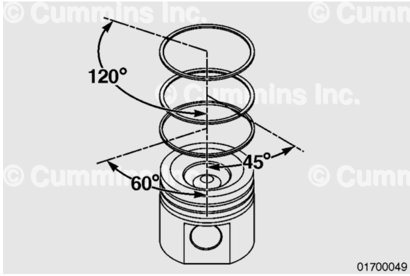
Make sure the tangs on the bearing shells are in the slot of the connecting rod cap and connecting rod.

Lubricate the connecting rod bearings with clean lubricating engine oil.



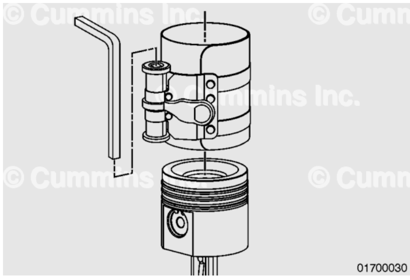
Position the piston ring gaps as shown in the illustration.

- Top Ring: 120° **clockwise** from the front of the piston.
- Second Ring: 45° **counterclockwise** from the front of the piston.
- Oil Ring: 60° **clockwise** from the front of the piston.



⚠ CAUTION ⚠

If using a strap type ring compressor, make sure the inside end of the strap does not hook on a ring gap and break the ring.

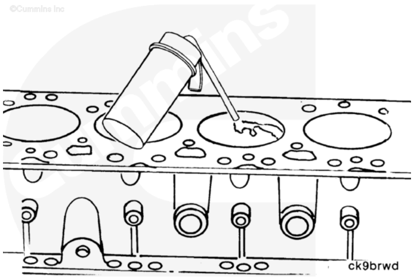


Use a piston ring compressor to compress the rings.

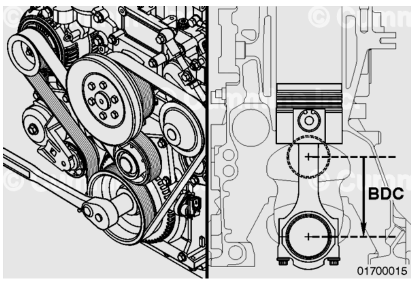
Lubricate the cylinder bore with clean 15W-40 engine lubricating oil.

The cylinder block **must** be clean before assembly.

Inspect the cylinder bores for reuse. Refer to Procedure 001-026 in Section 1. (/qs3/pubsys2/xml/en/procedures/441/441-001-026.html)

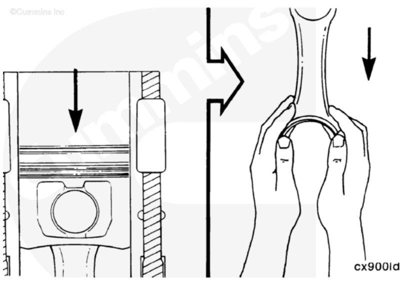


Position the connecting rod journal for the piston to be installed to bottom dead center (BDC).

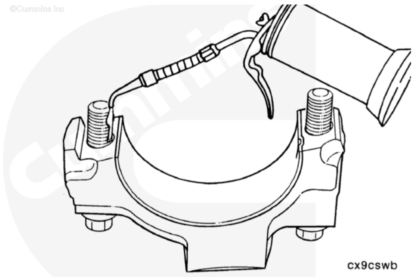


Take care to **not** damage the cylinder wall when inserting the connecting rod.

Carefully push the piston into the bore while guiding the connecting rod to the crankshaft journal.

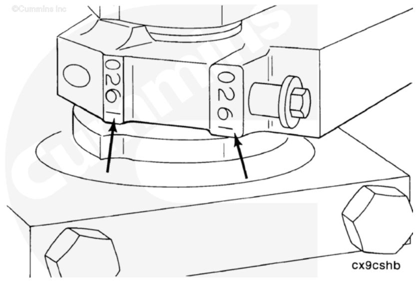


Lubricate the threads and underside of the connecting rod capscrew heads with clean 15W-40 lubricating engine oil.



⚠ CAUTION ⚠

The number stamped on the rod and cap at the parting line must match and be installed facing the exhaust side of the engine.



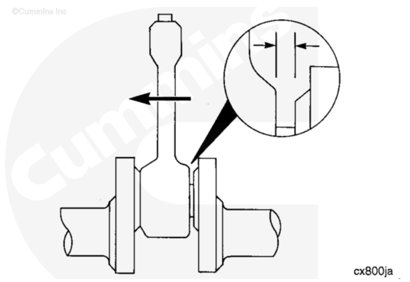
Install the connecting rod cap and capscrews.

Tighten the two capscrews in alternating sequence.

Torque Value:

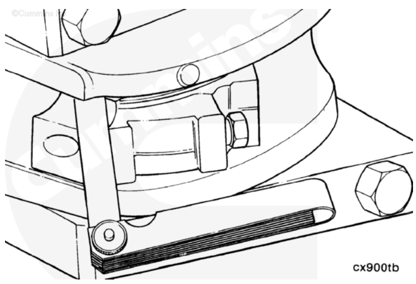
1. 39 n•m [29 ft-lb]
2. Rotate each capscrew 90 degrees.

Check the side clearance between the connecting rod and crankshaft. The connecting rod **must** move freely from side-to-side on the crankshaft journal. If the rod does **not** move freely, remove the rod cap and make sure the bearing shells are the correct size.



Note : Do not measure the clearance between the cap of the connecting rod and crankshaft.

Measure the side clearance between the connecting rod and crankshaft.



Connecting Rod Cap Side Clearance		
mm		in
0.20	MIN	0.0079
0.40	MAX	0.0160

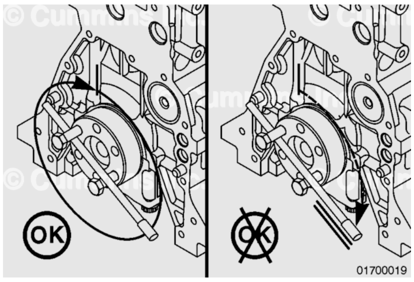
Repeat the above steps to install the remaining connecting rods and bearings.

⚠ CAUTION ⚠

To reduce the possibility of engine damage, the crankshaft must rotate freely.

⚠ CAUTION ⚠

If the connecting rod is not properly oriented (tang opposite the camshaft), it will contact the camshaft and lock the engine.



Check for freedom of rotation as the connecting rod caps are installed. If the crankshaft does **not** rotate freely, check the installation of the connecting rod bearings and bearing size.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/441/441-002-004.html)
- Install the lubricating oil pan and gasket. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/441/441-007-025.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/441/441-008-018-tr.html)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/441/441-007-037.html)
- Connect the batteries. See equipment manufacturer service information.

- Operate the engine to normal operating temperature.
  Check for leaks.